Prime Geometry I:
The Prime Triangle and its Geometric Invariants

Allen Proxmire

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Abstract

To each pair of consecutive primes (p_n, p_{n+1}) we associate a right triangle with legs p_n and p_{n+1}. The angle $\alpha_n = \arctan(p_n/p_{n+1})$ and its associated monotone proxy $\rho_n = p_n/p_{n+1}$ are the central geometric quantities; the hypotenuse $C_n = \sqrt{p_n^2 + p_{n+1}^2}$ and, for triples, the difference $C_2 - C_1$ generate a family of elementary invariants. We derive the Prime Square-Difference identity $C_2^2 - C_1^2 = p_{n+2}^2 - p_n^2$, define the integer-valued PSD factor, and record first-order approximations for the discrete energy E_n, curvature K_n, angle drift $\Delta\alpha_n$, and normalized shape curvature χ_n. We then introduce the angle-record sequence and state the $2P$-beats lemma that governs it, foreshadowing the dyadic-density interpretation of PG II and the Generalized Bertrand Principle of PG III. Finally we position this framework within the Ramanujan-prime family, from which the twin- and constellation-Ramanujan sequences arise as immediate analogs.

1 Introduction

Prime gaps exhibit rich structure that is invisible at the level of the primes alone. This paper introduces a minimal geometric construction which encodes consecutive primes as right triangles and records prime-gap dynamics in the associated angles. The construction is elementary, and most of its identities are algebraically routine. Its value lies not in depth but in *angle*: by shifting the viewpoint from gaps to triangles, one discovers quantities whose record structure encodes non-trivial density information about the primes.

The Prime Triangle framework can be thought of as a *geometric microscope* on prime gaps. Consecutive-prime pairs produce triangles whose angles cluster near 45°, with twin primes approaching the limit fastest. Asking which pairs set new records in the angle sequence turns out to be equivalent to asking about the density of twin primes in dyadic intervals — an observation developed fully in PG II and generalized in PG III. The present note establishes the geometric objects and identities on which those companion papers rest.

Three themes run through the paper:

(1) *Geometric encoding.* The triangle (p_n, p_{n+1}, C_n) records a prime pair; the angle $\alpha_n = \arctan(p_n/p_{n+1})$ records the asymmetry between the pair's members.

(2) *Derived quantities.* Elementary algebra yields an exact identity $C_2^2 - C_1^2 = p_{n+2}^2 - p_n^2$ and an always-integer quantity PSD_n. Gap-scaled approximations produce a discrete energy E_n, curvature K_n, and normalized shape curvature χ_n.

(3) *Dyadic interpretation.* The sequence of angle-records among consecutive prime pairs satisfies an elementary comparison rule that translates directly into a dyadic density statement on twin

primes and, more generally, on admissible prime constellations. This is the hinge between the geometric object (the Prime Triangle) and the arithmetic statements of PG II–III.

Throughout, $\mathbb{P}$ denotes the primes, $(p_n)_{n\geq 1}$ the ordered list of primes (so $p_1 = 2$, $p_2 = 3$, ...), and $\pi(x) = \#\{p \in \mathbb{P} : p \leq x\}$ the prime-counting function. For pair-constellation $\mathcal{C} = \{0, g\}$ with $g \geq 2$ even, the counting function $\pi_{\mathcal{C}}(x) = \#\{p \leq x : p \in \mathbb{P},\ p + g \in \mathbb{P}\}$ specializes to $\pi_2(x)$ when $g = 2$.

2 The Prime Triangle

2.1 Construction

Definition 1 (Prime Triangle)**.** *For consecutive primes $p_n < p_{n+1}$, the* Prime Triangle *at index n is the right triangle with legs p_n and p_{n+1} and hypotenuse*

$$C_n \ := \ \sqrt{p_n^2 + p_{n+1}^2}.$$

We record the pair's geometric parameters as

$$\alpha_n \ := \ \arctan\left(\frac{p_n}{p_{n+1}}\right), \qquad \rho_n \ := \ \frac{p_n}{p_{n+1}}.$$

The quantity α_n is the angle at the base of the leg of length p_{n+1}, and $\rho_n = \tan \alpha_n$.

Proposition 2 (Basic properties)**.** *For every $n \geq 1$:*

(i) $0 < \alpha_n < 45°$*, equivalently* $0 < \rho_n < 1$*.*

(ii) *ρ and α are strictly monotone: for pairs (p, q) and (p', q') with $p < q$ and $p' < q'$, $\rho(p, q) > \rho(p', q') \iff \alpha(p, q) > \alpha(p', q')$.*

(iii) *For a twin pair $(p, p + 2)$, $\rho(p, p + 2) = p/(p + 2)$ and $\alpha(p, p + 2) \to 45°$ as $p \to \infty$.*

(iv) *For any fixed gap $g \geq 2$, $\alpha(p, p + g) \to 45°$ as $p \to \infty$; twin pairs achieve this limit fastest, followed by cousin pairs, sexy pairs, and so on.*

Proof. (i) follows from $p_n < p_{n+1}$, giving $\rho_n < 1$ and hence $\alpha_n < 45°$. (ii) is immediate since arctan is strictly increasing on $\mathbb{R}_{>0}$. (iii) is direct computation: for twins, $\rho(p, p + 2) = 1 - 2/(p + 2) \to 1$. (iv) For gap g, $\rho(p, p + g) = 1 - g/(p + g)$; for fixed g and large p, $1 - \rho(p, p + g) \sim g/p$, so smaller g gives faster convergence. □

2.2 Numerical illustration

The first few Prime Triangles:

n	p_n	p_{n+1}	gap	α_n
1	2	3	1	33.690°
2	3	5	2	30.964°
3	5	7	2	35.538°
4	7	11	4	32.471°
5	11	13	2	40.236°
6	13	17	4	37.405°
7	17	19	2	41.820°
8	19	23	4	39.558°
9	23	29	6	38.428°
10	29	31	2	43.091°

Observe the behavior: twin pairs consistently produce the largest angles at each scale, with the angle monotonically approaching 45° along the twin subsequence. Non-twin pairs interleave between twin records and always fall short of the preceding twin angle. This empirical pattern is made precise by the $2P$-beats lemma (Section **??**) and the angle-record theorem of PG II.

3 Derived Quantities

The Prime Triangle construction extends naturally to triples of consecutive primes, producing the identities and derived invariants collected in this section. Throughout, fix three consecutive primes $p_n < p_{n+1} < p_{n+2}$ and write

$$C_1 \;=\; \sqrt{p_n^2 + p_{n+1}^2}, \qquad C_2 \;=\; \sqrt{p_{n+1}^2 + p_{n+2}^2}.$$

3.1 The Prime Square-Difference Identity

Theorem 3 (PSD identity)**.** *For every triple of consecutive primes,*

$$C_2^2 - C_1^2 \;=\; p_{n+2}^2 - p_n^2.$$

Equivalently, writing $G_n := p_{n+2} - p_n$ for the skip-one gap,

$$C_2^2 - C_1^2 \;=\; G_n\,(2p_n + G_n).$$

Proof. Direct expansion:

$$C_2^2 - C_1^2 = \left[p_{n+2}^2 + p_{n+1}^2\right] - \left[p_n^2 + p_{n+1}^2\right] = p_{n+2}^2 - p_n^2 = (p_{n+2} - p_n)(p_{n+2} + p_n) = G_n(2p_n + G_n). \quad \square$$

Definition 4 (PSD factor)**.** *The* Prime Square-Difference factor *at index n is*

$$\mathrm{PSD}_n \;:=\; \frac{p_{n+2}^2 - p_n^2}{12} \;=\; \frac{G_n\,(2p_n + G_n)}{12}.$$

Theorem 5 (Integrality and last-digit structure)**.** *For every n with $p_n \geq 5$:*

(i) $\mathrm{PSD}_n \in \mathbb{Z}$.

(ii) The last decimal digit of PSD_n *is one of* $\{0, 4, 6\}$.

Proof. Every prime $p \geq 5$ satisfies $p \equiv \pm 1 \pmod 6$. Writing $p_n = 6a \pm 1$, $p_{n+2} = 6b \pm 1$, a four-case expansion gives

$$(6b \pm 1)^2 - (6a \pm 1)^2 = 12(b-a)\big[3(a+b) \pm 1\big],$$

so $12 \mid p_{n+2}^2 - p_n^2$ and PSD_n is an integer. For (ii), reducing modulo $120 = \mathrm{lcm}(12, 10)$ gives $p_{n+2}^2 - p_n^2 \in \{0, 48, 72\} \pmod{120}$, so $\mathrm{PSD}_n \in \{0, 4, 6\} \pmod{10}$. □

Remark 6 (Twin-structure specialization)**.** *When $G_n = 6$, the three consecutive primes take one of the forms $(p_n, p_n + 2, p_n + 6)$ or $(p_n, p_n + 4, p_n + 6)$; in either case p_{n+1} is a member of a twin prime pair. A direct computation from the identity gives*

$$\mathrm{PSD}_n \;=\; \frac{6(2p_n + 6)}{12} \;=\; \frac{p_n + p_{n+2}}{2} \;=\; p_{n+1} \pm 1,$$

where the sign encodes which adjacent pair forms the twin. This is the simplest case in which PSD_n *pinpoints twin-prime structure within a triple.*

3.2 Energy, curvature, and angle drift

Definition 7 (Energy and skip-one gap)**.** *For three consecutive primes, define the* energy transition

$$E_n \;:=\; C_2 - C_1,$$

and recall $G_n = p_{n+2} - p_n = g_n + g_{n+1}$ with $g_n = p_{n+1} - p_n$.

Proposition 8 (First-order expansion of E_n)**.** *With $g_n, g_{n+1} \ll p_n$,*

$$E_n \;=\; \frac{\sqrt{2}}{2}\, G_n \;+\; O\!\left(\frac{G_n^2}{p_n}\right),$$

and more precisely

$$\frac{E_n}{G_n} \;\approx\; \frac{\sqrt{2}}{2} - \frac{g_n - g_{n+1}}{2\sqrt{2}\,(2p_n + G_n)}.$$

Proof sketch. Taylor-expand $\sqrt{(p_{n+1} + g_{n+1})^2 + p_{n+1}^2} - \sqrt{p_n^2 + p_{n+1}^2}$ in g_{n+1}/p_{n+1} and g_n/p_n and collect first-order terms. □

Definition 9 (Discrete curvature and shape curvature)**.**

$$K_n := E_{n+1} - E_n \qquad \textit{(discrete curvature)},$$
$$\chi_n := K_n / E_n \qquad \textit{(normalized shape curvature)}.$$

Proposition 10. *Under the same first-order approximation,*

$$K_n \;\approx\; \frac{\sqrt{2}}{2}\,(g_{n+2} - g_n), \qquad \chi_n \;\approx\; \frac{g_{n+2} - g_n}{g_n + g_{n+1}}.$$

The quantity χ_n is a dimensionless, scale-free measure of how the prime-gap sequence bends at index n. It will be the central object of PG II and PG III's empirical sections when examining sign-coherence and long-range structure; for the present paper it is enough to record its definition.

3.3 Angle drift

Definition 11 (Angle drift)**.**

$$\Delta\alpha_n \;:=\; \alpha_{n+1} - \alpha_n.$$

Proposition 12 (Derivative hierarchy)**.** *In the first-order approximation, $\Delta\alpha_n$ behaves as a discrete first derivative of the sequence $\{\alpha_n\}$, and χ_n as a normalized discrete second derivative. Schematically,*

$$\chi_n \;\longrightarrow\; \Delta\alpha_n \;\longrightarrow\; \alpha_n,$$

with each arrow indicating discrete integration.

The hierarchy explains, in an elementary way, why the angle sequence α_n exhibits long coherent phases and suppressed extremes: curvature χ_n is small on average, so $\Delta\alpha_n$ is well-controlled, so α_n drifts smoothly toward its 45° limit.

4 Angle Dynamics and Records

4.1 The angle-record sequence

Definition 13 (Angle-record)**.** *A consecutive prime pair (p_n, p_{n+1}) is an* angle-record *if $\rho_n > \rho_m$ for every $m < n$, equivalently $\alpha_n > \alpha_m$ for every $m < n$.*

By Proposition **??**, the angle-record sequence is an infinite subsequence of the consecutive-prime-pair sequence. Its first few terms are:

record #	p_n	p_{n+1}	α_n	type
1	2	3	33.690°	gap-1 initial pair
2	5	7	35.538°	twin
3	11	13	40.236°	twin
4	17	19	41.820°	twin
5	29	31	43.091°	twin
6	41	43	43.636°	twin
7	59	61	44.045°	twin
8	71	73	44.204°	twin
⋮	⋮	⋮	⋮	⋮

4.2 The $2P$-beats lemma

The key elementary observation governing the record structure is the following comparison rule.

Lemma 14 ($2P$-beats)**.** *Let $(p, p+g)$ and $(q, q+h)$ be prime pairs with $g, h \geq 2$. Then*

$$\rho(p, p+g) \;>\; \rho(q, q+h) \quad\Longleftrightarrow\quad h\,p \;>\; g\,q.$$

In particular, a gap-g pair at p beats a twin pair $(q, q+2)$ in angle if and only if $p > (g/2)\,q$; the binding case (smallest g larger than 2) is $g = 4$, requiring $p > 2q$.

Proof. $\rho(p, p+g) > \rho(q, q+h) \iff p(q+h) > q(p+g) \iff hp > gq.$ □

Corollary 15 (Preview of the angle-record theorem)**.** *Let $(T_k)_{k\geq 1}$ be the twin-prime subsequence (smaller members). Suppose that for every k with $T_k \geq 11$, $T_{k+1} < 2T_k$. Then every angle-record with leading prime $p_n \geq 3$ is a twin prime.*

Proof sketch. If a non-twin pair $(p, p+g)$ with $g \geq 4$ were to set a record after the twin T_k and before T_{k+1}, Lemma **??** would require $p > (g/2)T_k \geq 2T_k$. But $p \leq T_{k+1} < 2T_k$ by hypothesis; contradiction. □

The converse direction and the full equivalence with the dyadic density inequality $\pi_2(2x) - \pi_2(x) \geq 1$ are developed in PG II.

5 Geometric Interpretation of Dyadic Density

The $2P$-beats lemma reveals a direct equivalence between angle dynamics and dyadic density. Corollary **??** asserts that the factor-2 bound on twin-prime ratios $T_{k+1} < 2T_k$ is exactly what guarantees twins dominate the angle-record sequence. Conversely, if the angle-record sequence omits a twin prime (i.e., some non-twin pair sets a record), then some ratio $T_{k+1}/T_k \geq 2$ must occur.

5.1 Bertrand factors, revisited geometrically

The dyadic factor 2 is, in classical terms, Bertrand's factor: every integer $n > 1$ has a prime in $(n, 2n]$. For twin primes, the analogous dyadic statement

$$\pi_2(2x) - \pi_2(x) \;\geq\; 1 \qquad (x \geq 11),$$

which we formalize as the Twin-Prime Bertrand Postulate (TPB) in PG II, is exactly the ratio condition $T_{k+1} < 2T_k$ for $T_k \geq 11$ via a direct one-line argument.

Geometrically, Lemma **??** explains the recurrence of the factor 2: for a gap-g constellation $\mathcal{C} = \{0, g\}$ and its *doubled companion* $\mathcal{C}' = \{0, 2g\}$, a $\mathcal{C}'$-pair at P beats a $\mathcal{C}$-pair at Q in angle if and only if $P > 2Q$ — independent of g. The dyadic factor 2 is therefore also the universal angle-beats factor between doubled constellations, providing a geometric rationale for the uniform Bertrand factor that appears in PG III's Generalized Bertrand Principle.

5.2 A summary of the geometric–arithmetic correspondence

We can summarize the correspondence in one line:

Angle-records dominated by a gap-g constellation $\iff$ *dyadic density $\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \geq 1$ holds for all x past a threshold.*

For $g = 2$ (twins), the dominance is empirically total once we exclude the initial non-twin $(2, 3)$; the equivalent dyadic inequality is (TPB). For larger g the statement is modified by the role of smaller-gap competitors (twins beat cousins at angle almost everywhere), but the *within-constellation* dyadic inequality

$$\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \;\geq\; 1$$

persists with the same Bertrand factor of 2. The full picture is developed in PG III as the Generalized Bertrand Principle.

6 The Ramanujan-Prime Context

Bertrand's postulate $\pi(2x) - \pi(x) \geq 1$ for $x \geq 1$, proved by Chebyshev in 1852, is the $n = 1$ case of a broader family. Ramanujan [**?**] proved in 1919 that for every integer $n \geq 1$, the inequality $\pi(x) - \pi(x/2) \geq n$ holds for all sufficiently large x; the n-th *Ramanujan prime* is defined as

$$R_n \;=\; \min\Big\{\, R \,:\, \pi(x) - \pi(x/2) \geq n \text{ for all } x \geq R \,\Big\}, \tag{1}$$

so $R_1 = 2$. The first several values are $R_n = 2, 11, 17, 29, 41, 59, 67, 71, \ldots$ (OEIS `A104272` [**?**]). Sondow [**?**] established the asymptotic $R_n \sim p_{2n}$ and initiated the modern study of this sequence; later work [**?**, **?**] introduced c-Ramanujan primes (for interval ratios other than 1/2) and derived Ramanujan-prime iterations.

The Prime Triangle framework makes two direct analogs transparent:

- *Twin-Ramanujan primes.* Replacing π by π_2 in (**??**) gives the sequence

  $$R_n^{\text{twin}} \;=\; \min\Big\{\, R \,:\, \pi_2(x) - \pi_2(x/2) \geq n \text{ for all } x \geq R \,\Big\}.$$

  Here R_1^{twin} is exactly the Twin-Prime Bertrand threshold (TPB), equal to 11. Higher values R_n^{twin} quantify the higher-order dyadic density of twin primes.

- *Constellation-Ramanujan primes.* For each admissible $\mathcal{C} = \{0, g\}$ the sequence

  $$R_n^{\mathcal{C}} \;=\; \min\Big\{\, R \,:\, \pi_{\mathcal{C}}(x) - \pi_{\mathcal{C}}(x/2) \geq n \text{ for all } x \geq R \,\Big\}$$

  is the natural extension; the cases $g = 4, 6$ give $R_1^{\{0,4\}} = 7$ and $R_1^{\{0,6\}} = 5$.

Each of these sequences is empirically a subsequence of its defining constellation (every R_n^{twin} is a twin prime, every $R_n^{\mathcal{C}}$ is a member of $\mathcal{C}$), because the relevant counting function jumps upward only at constellation members. PG II develops R_n^{twin} explicitly and PG III generalizes to other $\mathcal{C}$; the present paper provides the geometric setting in which these are natural.

7 Conclusion

The Prime Triangle construction is elementary. Its value is in how it organizes. The angle α_n is monotone-equivalent to the ratio $\rho_n = p_n/p_{n+1}$, both of which record the geometric shape of consecutive primes. The triangle's hypotenuse difference $C_2 - C_1$ yields an exact identity with the skip-one square difference $p_{n+2}^2 - p_n^2$, producing the integer-valued PSD_n factor. First-order expansions give an energy E_n, curvature K_n, normalized shape curvature χ_n, and angle drift $\Delta\alpha_n$ that form a derivative hierarchy.

These objects are the geometric setting for the main conjectural content of the PG series:

- The sequence of angle-records selects, via the $2P$-beats lemma, a subfamily of consecutive-prime pairs that is equivalent (up to the trivial $(2, 3)$) to the twin primes. This equivalence is the Angle-Record Theorem of PG II.

- The equivalence extends via the same $2P$-beats threshold to a dyadic density statement: at least one twin prime in every $(x, 2x]$ with x past an effective threshold. This is the Twin-Prime Bertrand Postulate of PG II, with threshold $R_1^{\text{twin}} = 11$.

- Generalization to admissible pair-constellations yields the Generalized Bertrand Principle of PG III, and a corresponding family of constellation-Ramanujan primes $R_n^{\mathcal{C}}$ that extends the Ramanujan-prime framework beyond ordinary primes for the first time.

The Prime Triangle is thus a small object with a long reach: from an elementary algebraic identity through an elementary geometric comparison lemma to an explicit, unproven, well-posed conjectural statement on prime-constellation density. The arithmetic consequences are developed in the companion papers.

Acknowledgements

This paper is the first in a three-part series. PG II introduces the Twin-Prime Bertrand Postulate and proves the Angle-Record Theorem. PG III generalizes to admissible pair-constellations and establishes the Generalized Bertrand Principle, including the twin- and constellation-Ramanujan prime sequences.

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